

Multiple Choice (10 marks)

1. What is the evidence for the theory that the neurilemma sheath plays a role in the regeneration of severed nerves?
 - (a) The neurilemma sheath absorbs toxins at the injury site to facilitate regeneration.
 - (b) The neurilemma sheath provides a channel for the axon to grow back to its former position.
 - (c) The neurilemma sheath fuses severed axon ends together.
 - (d) The neurilemma sheath repels inhibitory molecules that prevent nerve growth.
 - (e) The neurilemma sheath acts as a source of neural stem cells for regeneration.
2. Why would one call a flock of birds a "society"?
 - (a) Birds are called a society because each bird takes a turn leading the flock.
 - (b) Birds in a flock are unrelated
 - (c) A flock of birds can be considered an elementary form of society called a motion group.
 - (d) The collective intelligence of a flock is referred to as a society.
 - (e) Birds form a society to perform individual activities
3. All of the following statements about plant embryogenesis are correct EXCEPT:
 - (a) The suspensor is derived from the basal cell.
 - (b) Cotyledons are derived from the apical cell.
 - (c) Shoot apical meristem formation occurs after seed formation.
 - (d) Precursors of all three plant tissue systems are formed during embryogenesis.
4. Why is the organism that causes smallpox not used in the vaccine against smallpox?
 - (a) The organism is extinct in the wild and cannot be harvested for vaccine production
 - (b) The organism that causes smallpox is too costly to use in vaccine production
 - (c) The use of the actual virulent variola virus may be dangerous if too much is administered
 - (d) Smallpox virus is confined to research laboratories
 - (e) There is no specific treatment for smallpox
5. Convergent evolution is best exemplified by which of the following?
 - (a) The gills of a fish and the lungs of a human

- (b) The leaves of an oak tree and the spines of a cactus
- (c) The eyes of a human and the eyes of a squid
- (d) The roots of a tree and the legs of an animal
- (e) The presence of a notochord in all chordate embryos

True / False (10 marks)

Answer True or False

- 6. True or False: Does oral -galactosidase relieve irritable bowel symptoms?
- 7. True or False: Does maternal obesity have an influence on feeding behavior of obese children?
- 8. True or False: Serum angiotensin-converting enzyme and frequency of severe hypoglycaemia in Type 1 diabetes: does a relationship exist?
- 9. True or False: Should general practitioners call patients by their first names?
- 10. True or False: Is withdrawal-induced anxiety in alcoholism based on beta-endorphin deficiency?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the significance of stem reptiles in evolutionary biology, focusing on their unique evolutionary trajectory and how they exclusively led to the emergence of modern reptiles.
- 12. Discuss how the mapping of vegetation formations or biomes can enhance a climatologist's understanding of the climatic conditions within the biosphere.

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